



Finish the Stack

Quick facilitation notes & answer key

Goal

Children complete a partial block program by choosing the one move block that makes Bit reach the star, then run the whole stack to confirm. This is early program design: picking the instruction that produces the intended result, and testing it.

Kindergarten frame (Ontario)

Problem Solving and Innovating — children select and test instructions to reach a goal.

No reading required of the child: read each prompt aloud; the child circles ➡ or ⬅ and draws the arrow in the empty block.

Materials

The printed worksheet, a pencil, and a blue crayon. Optional: floor boxes 1-5 with a movable star so the child can test the chosen block.

Run it (about 15 min)

1. MODEL: on Program A, run ➡ ➡ together to box 3. 'Bit is on box 3 but the star is box 4 — one more step right. The missing block is ➡.'
2. GUIDED: Exercise 1 — children circle ➡ and draw it. Exercise 2 — the gap is in the middle, but the answer is still ➡; run the whole stack to confirm box 4.
3. YOU TRY: Exercise 3 — the star is now to the LEFT (box 2), so the missing block is ⬅.
4. Connect: 'Programmers add the block that finishes the job — then they run it to make sure.'

Answer key (modelled by the run-gate)

Ex 1 — Program A: missing block is ➡ (box 1 → 2 → 3 → 4, the star).

Ex 2 — Program B: missing block is ➡, in the middle (box 1 → 2 → 3 → 4).

Ex 3 — Program C: missing block is ⬅ (box 4 → 3 → 2, the star).

Watch for & differentiate

Common trap: picking ⬅ in Ex 1-2 because it is offered — have the child run the stack and see Bit move away from the star.

Ex 3 reverses the direction; re-anchor 'the star is on the LEFT now, so Bit goes ⬅.'

Simplify: do Programs A and C only (clear right / clear left).

Extend: ask what the missing block would be if the star moved to box 5 in Stage 1 (still ➡, and Program A would need two more steps).

Movement option: walk the fixed blocks on the floor, then let the child choose the last step to land on the star.

Success looks like

The child draws ➡ for Programs A and B and ⬅ for Program C, and can run each completed stack to show Bit landing on the star.